

Chapter 1 Introduction

1.1 Background and Functional Split Evolution

To address the diverse and stringent requirements of 5G networks, the 3rd Generation Partnership Project (3GPP) explored various functional split options to disaggregate the traditional monolithic base station architecture [?]. Building upon this foundation, the O-RAN Alliance further standardized the disaggregated architecture into three primary logical nodes: the Open Central Unit (O-CU), Open Distributed Unit (O-DU), and Open Radio Unit (O-RU). This disaggregation allows operators to optimize transport bandwidth and computational resources based on their specific deployment environments.



Currently, the industry recognizes three mainstream functional split configurations. The first is **Split Option 2**, which defines the interface between the O-CU and O-DU. The second is **Split Option 7.2**, a low-layer split standardized by O-RAN between the O-DU and O-RU [?]. The third is **Split Option 6**, a MAC-PHY split (often referred to as the O-DU High/Low split), which is the primary focus of this thesis. To ensure interoperability for Option 6, the Small Cell Forum (SCF) has unified the interface specifications under the network Functional Application Platform Interface (nFAPI) [?].

These three mainstream split options exhibit fundamental differences in their synchronization requirements and management mechanisms:

- **Split Option 2 (CU/DU Split):** The split point is located between the PDCP and RLC layers. Because it operates at a higher protocol layer, its synchronization requirement is based on **Sequence Numbers** rather than strict time domains. The core management mechanism relies on **Flow control and Deep Buffering** to handle packet delivery. Consequently, it is well-suited for deployment in **General packet-switched networks** without requiring precise physical-layer timing alignment.
- **Split Option 6 (DU High/Low split):** Operating between the MAC and PHY layers, this split requires a more stringent **Frame/Slot** level of synchronization. Its core management focuses heavily on **Delay management** to ensure scheduling instructions arrive before their intended slot boundaries. Like Option 2, it is designed to operate over **General packet-switched networks**, but it must adaptively adjust to network jitter.
- **Split Option 7.2 (DU/RU Split):** Located within the physical layer, this split demands extreme **Phase/Symbol** level synchronization. Its core management requires strict hardware-assisted mechanisms such as **Hardware PTP and Window** enforcement. Due to its high sensitivity to delay and jitter, it necessitates deployment over specialized **Time-Sensitive Networking (TSN)** environments.

Table 1.1 summarizes the technical characteristics of the three split architectures regarding synchronization maintenance.

This architectural difference means that when facing non-ideal fronthaul networks, Option 6 exhibits a better delay tolerance compared to Op-

Table 1.1: Comparison of Functional Splits and Timing Maintenance

Split Option	Sync Level	Core Management	Deployment Environment
Option 2 (CU/DU Split)	Sequence Number	Flow control & Deep Buffering	General packet-switched networks
Option 6 (DU High/Low split)	Frame/Slot	Delay management	General packet-switched networks
Option 7.2 (DU/RU Split)	Phase/Symbol	Hardware PTP & Window	Time-Sensitive Networking

tion 7.2, while providing more granular physical-layer control compared to Option 2.



1.2 Motivation and Problem Statement

The necessity of investigating high latency in Lower Layer Split (LLS) architectures is highlighted in the O-RAN Fronthaul working group specification [?]. Specifically, Annex L of the specification discusses considerations for operating in non-ideal transport environments. Long latency fronthaul links may break traditional Hybrid ARQ (HARQ) timelines, leaving only a tight 4 ms processing budget for the base station. This constraint often necessitates the use of real-time operating systems (RTOS) and dedicated hardware. However, for low- to medium-impact deployment scenarios, such as indoor small cells and fixed wireless access, the performance impact of delayed HARQ or predictive HARQ is negligible [?]. Thus, en-

forcing sub-millisecond latency constraints restricts the exploration of innovative, cost-effective deployments.

In this context, the nFAPI (Split Option 6) architecture inherently aligns with the requirements of non-ideal fronthaul deployments. By separating the MAC and PHY layers, Split 6 exhibits a much higher latency tolerance (typically 2 ms to 10 ms) compared to Split 7.2x. This enhanced tolerance allows the Virtualized Network Function (VNF) hosting the MAC layer to be deployed on General Purpose Processors (GPPs) in cloud environments, eliminating the strict requirement for expensive dedicated fiber and precise synchronization hardware. Furthermore, nFAPI incorporates timing procedures that enable the MAC layer to adjust its scheduling behavior based on fronthaul latency conditions, directly addressing the issue of broken HARQ windows.



The rationale for migrating the MAC layer to a centralized cloud environment centers on the concept of resource pooling and elasticity. In 5G vRAN architectures, the computational load of the physical layer (L1) is primarily determined by relatively static parameters, such as system bandwidth and antenna configuration. In contrast, the processing complexity of the Layer 2 (L2) stack, particularly the MAC scheduler, exhibits a linear or even quadratic dependency on the number of active User Equipments (UEs) and traffic volume [?,?]. For instance, as the number of connected UEs increases, the MAC scheduler must continuously evaluate proportional fair metrics, process buffer status reports, and handle logical channels, leading to a substantial surge in CPU cycles [?]. By decoupling the MAC layer into a VNF, the cloud infrastructure can dynamically allocate virtual cores to accommodate fluctuating UE densities, maintaining cost-effective scal-

ing compared to local PNF deployments.

However, this resource scaling introduces a new challenge for synchronization. As the number of UEs increases, the MAC processing time ($T_{\text{process_VNF}}$) inevitably lengthens within a given Transmission Time Interval (TTI). This extended processing duration compresses the remaining time margin available for fronthaul transmission latency. If the scheduling mechanism does not account for this endogenously increasing processing delay, static timing configurations will result in scheduling messages arriving at the PNF too late. Although Option 6 possesses flexible timing adjustment capabilities, its dynamic adjustment algorithms remain immature in existing open-source implementations, such as OpenAirInterface (OAI). Consequently, when confronted with non-deterministic operating system scheduling, UE-dependent processing load, and network congestion, the system frequently encounters “Missing Subframe” errors. This thesis selects Split 6 as the primary research subject, proposing an Adaptive Advance-Time Control architecture to dynamically compensate for both processing and transport latency variations, effectively overcoming these synchronization failures. A detailed analysis of this non-deterministic latency and the corresponding verification of these problems will be presented in Chapter ??, specifically isolating these issues in Scenario I (??).

1.3 Related Works

The evolution of Radio Access Network (RAN) architectures from Distributed RAN (D-RAN) to Open RAN (O-RAN) requires a thorough study of functional splits [?, ?, ?]. A fundamental challenge in deploying disag-

gregated RAN over non-ideal transport networks is managing the impact of fronthaul latency and functional split constraints. The O-RAN architecture introduces various functional splits to adapt to different network environments, each posing distinct synchronization requirements. For instance, recent work on cell-free massive MIMO (CF-mMIMO) in O-RAN analyzes hybrid fronthaul planning under functional splits 7.2x and 8, highlighting how fronthaul capacity and processing placement impact overall cost and performance in dense deployments [?]. In these 7.2x-based architectures, the stringent fronthaul timing requirements imply that any packets arriving outside the RU timing window due to jitter are passively discarded (i.e., out-of-window packet discard), making the system highly sensitive to non-ideal transport conditions. In contrast, our design adopts an Option 6 MAC-PHY split based on nFAPI, which enables microsecond-level, active phase and timing adjustments at the MAC-PHY boundary. This allows the system to partially absorb fronthaul jitter by proactively aligning transmission timing, rather than passively dropping late packets as in the conventional Option 7.2x fronthaul processing chain.

Current research focuses extensively on optimizing resource allocation and latency management in virtualized and cloud RANs. As summarized in Table 1.2, recent studies at top-tier venues address latency from various architectural levels. For example, Adamuz-Hinojosa *et al.* [?] proposed ORANUS, a latency-tailored orchestration framework for 6G O-RAN. While their work effectively bounds ultra-reliable low-latency communication (URLLC) queues using stochastic network calculus on a timescale of 10 ms to 1 s, it operates at the higher orchestration level rather than addressing microsecond-level synchronization.

Furthermore, virtualizing the MAC scheduler introduces significant server processing variations. Lv *et al.* [?] addressed this by jointly optimizing scheduling and orchestration to guarantee QoS in 5G vRANs. Their study highlights that shifting MAC processing to cloud VNF servers introduces endogenous processing latency, which compresses the permissible fronthaul transmission delay margin. However, their optimization predominantly targets computational resource contention rather than dynamically adapting to transport network jitter in a strict slot-level synchronization context like Split Option 6.

The necessity for highly strict, microsecond-level timing control in disaggregated architectures is further demonstrated by Chen *et al.* [?], who designed a sub-millisecond scheduler for multi-cell MIMO networks under the C-RAN architecture. Their work proves the severe constraints of sub-millisecond scheduling, illustrating that static timing advance configurations—commonly used in existing open-source frameworks like OpenAir-Interface (OAI)—are highly susceptible to missing subframe errors under network jitter. This highlights the critical gap filled by our research: while existing top-tier literature focuses on high-level orchestration or centralized baseband processing, our thesis specifically addresses the microsecond phase-level synchronization challenges inherent to Option 6. We propose a dynamic, dual-loop Adaptive Advance-Time Control algorithm and validate it on an end-to-end commercial testbed (COTS UE and Pegatron O-RU), proving its efficacy against non-deterministic fronthaul jitter and system variations.

1.4 Contributions

- **Delay Management in nFAPI Split 6:** We analyze and quantify the non-deterministic latency factors, including network jitter and server processing time, that lead to synchronization failures in disaggregated RAN deployments.
- **Adaptive Advance-Time Control Algorithm:** We propose a dynamic timing adjustment algorithm using a dual-loop architecture. The Convergence Optimization Loop estimates end-to-end jitter, while the VNF Timing Actuator adjusts scheduling lead times to ensure punctual delivery of P7 messages.
- **Experimental Validation:** We prototype the proposed mechanism on the OAI platform. Experimental results show that our algorithm eliminates “missing subframe” errors and maintains stable throughput under non-ideal fronthaul environments.

The rest of the thesis is organized as follows. Chapter ?? defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Chapter ??. Chapter ?? shows the numerical results. Concluding remarks are given in Chapter ??.

Table 1.2: Comparative Analysis of Related Works

Paper	Architecture & Split	Target Latency Issue	Time Scale	Co-Evaluation Method
[?] TWC 2026	O-RAN (FS 7.2x & 8)	Fronthaul capacity & cost trade-off	Static planning	Neural network ops simulation
[?] INFOCOM 2024	O-RAN (RIC-based)	URLLC queuing & transmission delay	10ms - 1s	Reinforcement learning simulation via SNO
[?] INFOCOM 2022	C-RAN (BBU-RRH)	Centralized base-band processing delay	< 1ms	GPU/GTS parallel joint scheduling
[?] INFOCOM 2024	5G vRAN (CU-DU)	Server non-deterministic	Cross-timescale	Joint AI scheduling